Tech Assessment: Weather Trend Forecasting

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PM Accelerator Mission: By making industry-leading tools and education available to individuals from all backgrounds, we level the playing field for future PM leaders.

1. Data Loading & Time-Series Structuring

1.1 Objective

The first step was to load the Global Weather Repository dataset and prepare it for time-series forecasting. Since the task requires using the *last_updated_epoch* and *last_updated* feature for modeling, building a clean and reliable time structure was essential before any analysis.

1.2 Data Ingestion

The dataset was loaded from the project's local data directory using pandas. Relative paths were used instead of absolute system paths to ensure the project runs correctly when cloned on another machine.

This keeps the repository portable and reproducible.

1.3 Column Standardization

To avoid inconsistencies and referencing errors, all column names were standardized:

- Converted to lowercase
- Removed extra spaces
- Replaced spaces and hyphens with underscores

This ensures cleaner downstream preprocessing and modeling.

1.4 Time Variable Selection

The dataset includes two time-related columns:

- *last_updated* (local time)
- *last_updated_epoch* (Unix timestamp)

For forecasting, *last_updated_epoch* was selected because it represents absolute UTC time. This avoids timezone-related inconsistencies and guarantees correct chronological ordering across all locations.

The Unix timestamp was converted into a proper datetime format, enabling time-based indexing, sorting, and modeling.

1.5 Time-Series Index Construction

Weather conditions vary by location, so each city must be treated as an independent time series.

A multi-index was created using:

(location_name, country, last_updated_epoch)

This structure allows for the separation of forecasting per city and proper chronological sorting. It will also allow time-based train/test splits and will maintain temporal integrity.

2. Data Cleaning & Preprocessing

2.1 Missing Value Assessment

The dataset was first evaluated for missing values using a column-wise summary of missing counts and percentages. A systematic check confirmed that no missing values were present across the dataset.

As a result, no imputation or row removal was required. This ensured that the temporal integrity of the dataset remained intact and no artificial bias was introduced during preprocessing.

2.2 Outlier Detection

Although no missing values were detected, extreme observations were examined using a robust statistical approach.

Outliers were identified using a median-based z-score (MAD method). Unlike standard z-scores, which rely on mean and standard deviation, the robust z-score uses:

- Median as the center
- Median Absolute Deviation (MAD) as the spread

This method is more suitable for environmental and meteorological data, which often contain heavy-tailed distributions and legitimate extreme events.

A threshold of $|z| > 3.5$ was used, following established statistical recommendations for robust anomaly detection.

Importantly:

- Outliers were flagged, not removed.
- Flag columns were added for each selected continuous variable.
- The original measurements were preserved.

This approach ensures that rare but real environmental events (e.g., heatwaves, storms, pollution spikes) remain part of the dataset while still being identifiable for further analysis.

2.3 Outlier Analysis

After flagging anomalies, the frequency of outliers was analyzed across variables. This allowed identification of which meteorological or air quality features exhibited the most extreme behavior.

Outlier analysis was applied only to continuous numerical measurement variables. Identifier fields (e.g., country, location_name), categorical columns (e.g., condition_text, wind_direction, moon_phase), and timestamp fields were excluded, as they do not represent measurable environmental values.

Outlier analysis showed that air quality variables had noticeably higher numbers of flagged observations compared to other weather features. For example, sulphur dioxide had about 19.6% of values flagged as extreme, and nitrogen dioxide had about 15.4%.

This is not surprising. Air pollution data often contains spikes due to traffic, industrial activity, weather conditions, or seasonal effects. These spikes are part of real-world environmental behavior and do not necessarily indicate measurement errors.

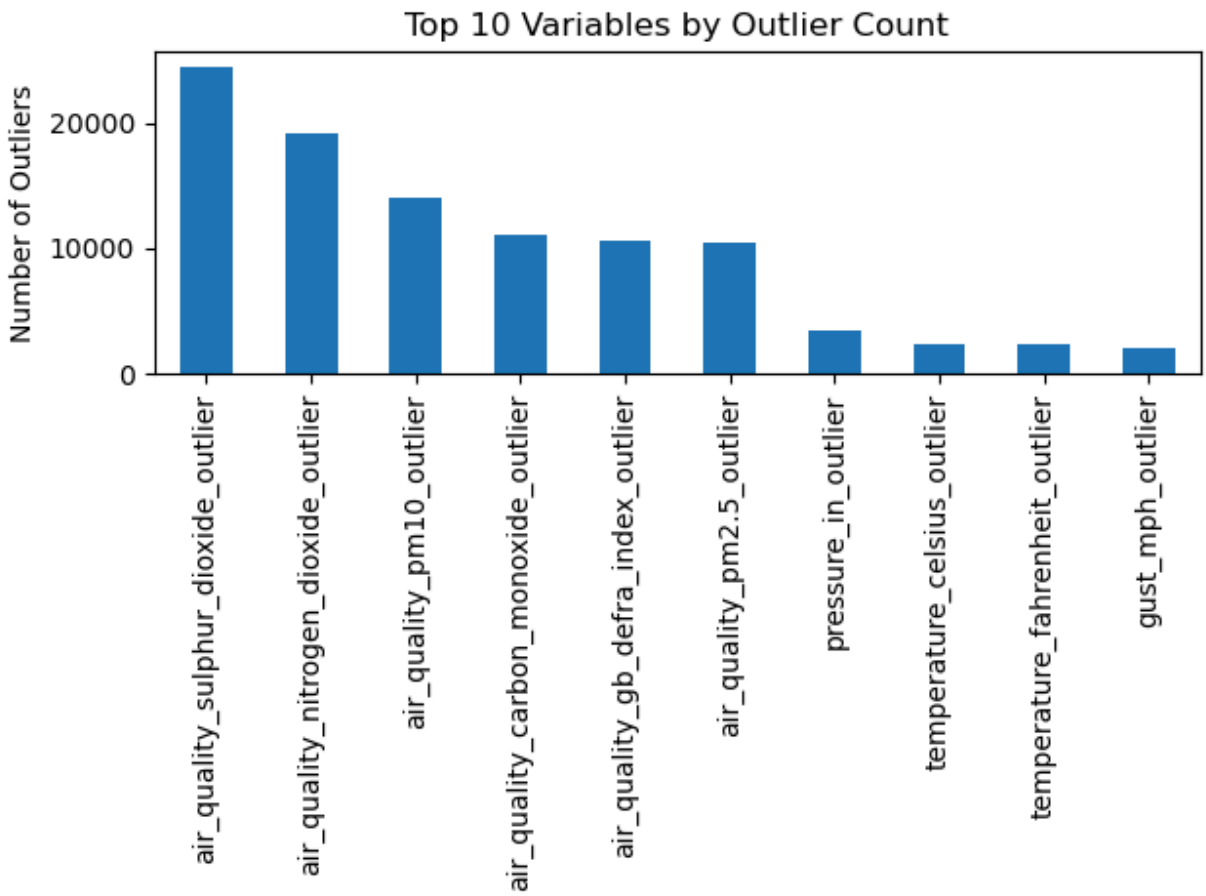
Because of this, the flagged values were not removed. Instead, they were kept in the dataset and treated as meaningful extreme environmental events. This helps preserve the true variability of air quality patterns across different locations.

The top 5 columns that have the highest number of outliers:

Columns	Outliers Numbers	Outliers Percentage
air_quality_sulphur_dioxide_outlier	24487	~19.6%
air_quality_nitrogen_dioxide_outlier	19253	~15.4%
air_quality_pm10_outlier	14025	~11.2%
air_quality_carbon_monoxide_outlier	11025	~8.8%

air_quality_gb_defra_index_outlier	10655	~8.5%

Visualization (Bar plots for top 10):



2.4 Feature Scaling

For modeling purposes, continuous variables were scaled using a **RobustScaler**, which:

- Centers values using the median
- Scales using the interquartile range (IQR)

This method is particularly appropriate for weather data because it reduces the influence of extreme values without distorting their relative importance.

Scaling was also applied only to continuous numerical measurement variables as outlier analysis.

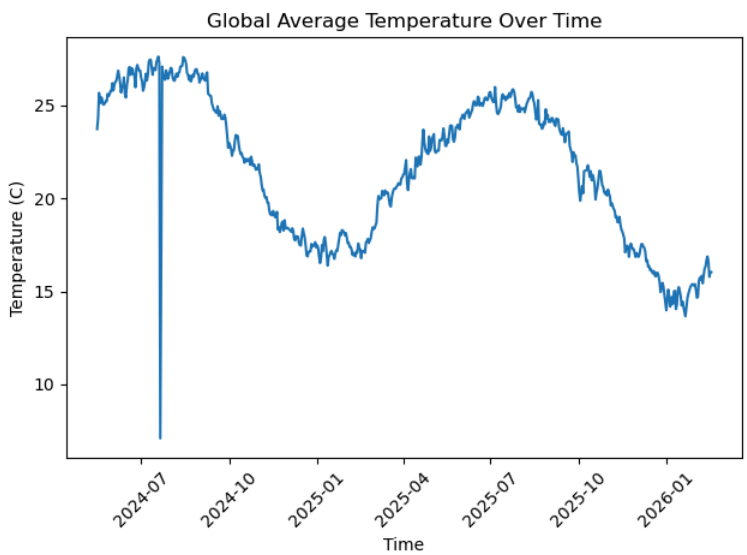
3. Exploratory Data Analysis (EDA)

3.1 Global Temperature Trend

To understand the overall climate pattern, the average temperature across all locations was calculated over time. Since multiple updates can occur within the same day, temperatures were aggregated at the daily level to reduce short-term fluctuations.

The resulting time-series plot shows a clear seasonal pattern. Global average temperatures increase during mid-year and decrease toward the end of the year. This repeating rise and fall suggests strong yearly seasonality, which supports the use of seasonal forecasting models in later stages.

One noticeable sharp drop appears in the early part of the timeline. This sudden decline is unlikely to represent a real global climate event. It is more likely caused by uneven data reporting or temporary differences in location coverage on that specific date. Aside from this isolated irregular point, the overall trend follows a consistent and realistic seasonal structure.



3.2 Temperature: City-Level Analysis

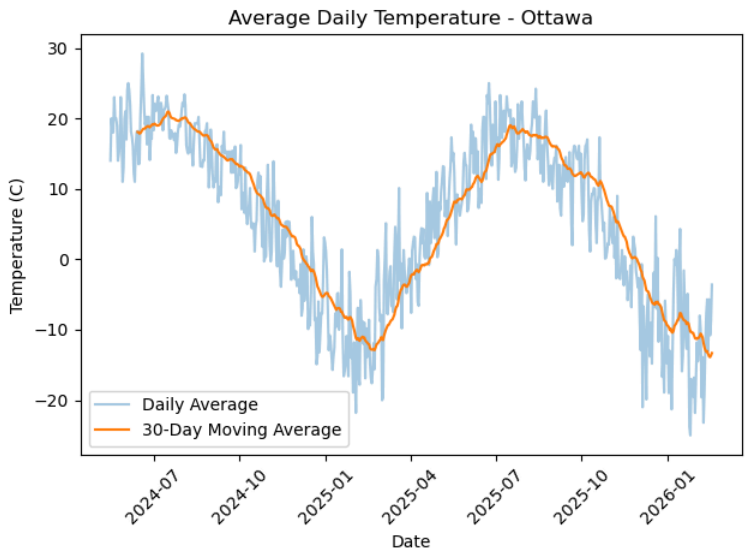
To better understand regional differences, daily average temperatures were analyzed for selected cities across different countries. The cities include Ottawa (Canada), Dhaka (Bangladesh), London (United Kingdom), Harare (Zimbabwe), and Canberra (Australia).

Ottawa (Canada)

The temperature trend for Ottawa shows a clear seasonal pattern:

- Warm summer months
- Very cold winters
- A smooth yearly rise and fall

A 30-day moving average was applied to reduce daily fluctuations and highlight the overall seasonal cycle. The smoothed curve clearly shows strong annual seasonality, with winter temperatures dropping below −10°C and summer temperatures reaching around 20°C.

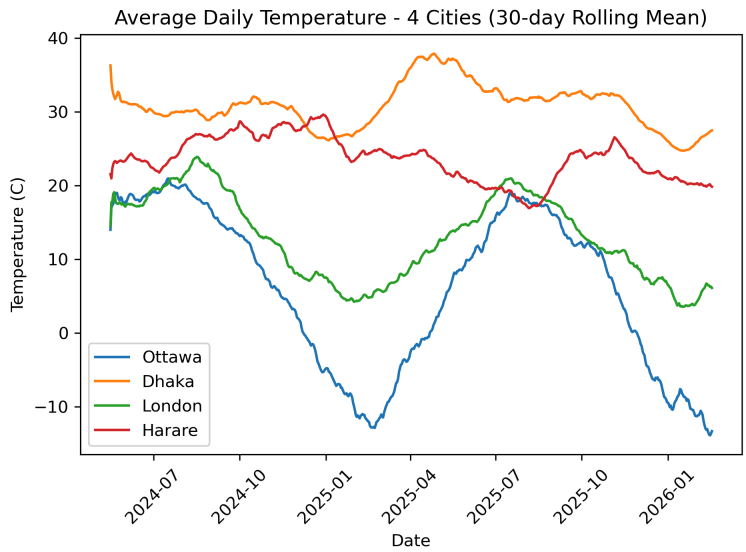


Comparison Across Four Cities

When comparing Ottawa, Dhaka, London, and Harare:

- Ottawa shows large seasonal variation, with strong winter and summer contrast.
- London shows moderate seasonal changes.
- Dhaka remains warm throughout the year, with smaller seasonal variation.
- Harare shows mild fluctuations with less extreme temperature changes.

This comparison highlights how climate patterns differ significantly across regions. Cities in colder climates show larger seasonal swings, while tropical or subtropical cities show more stable temperature patterns.



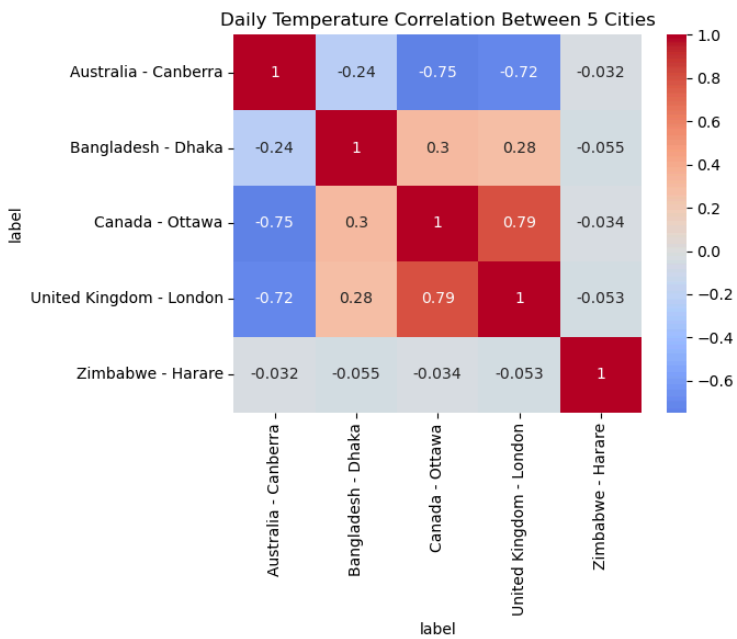
Correlation Between Five Cities

A correlation heatmap was also generated to measure how similar temperature patterns are between the selected cities.

Key observations:

- Ottawa and London show strong positive correlation (~0.79), meaning their seasonal patterns are similar.
- Canberra shows strong negative correlation with Ottawa and London (~−0.75), which is expected because Australia is in the Southern Hemisphere, where seasons are opposite.
- Dhaka shows weak to moderate positive correlation with some cities.
- Harare shows very weak correlation with the others, suggesting a different seasonal pattern.

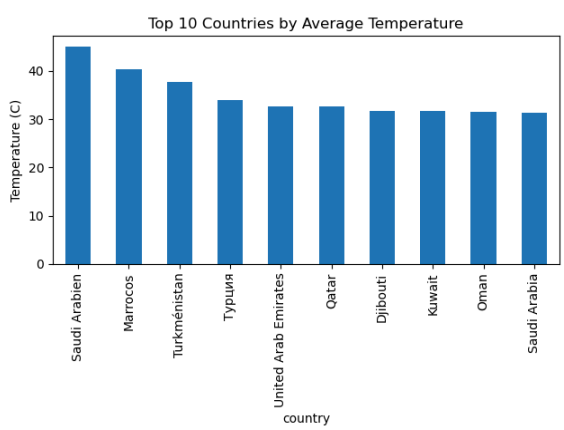
These results reflect real-world climate behavior and confirm that geographic location and hemisphere strongly influence temperature patterns.

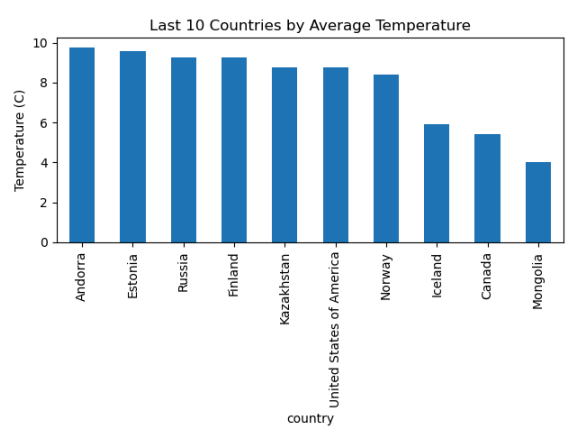


Overall, the city-level analysis confirms:

- Strong yearly seasonality in colder regions
- Opposite seasonal cycles between Northern and Southern Hemisphere cities
- Different levels of temperature variability depending on climate zone

This strengthens the case for using seasonal time-series models in forecasting.





3.3 Precipitation: City-Level Analysis

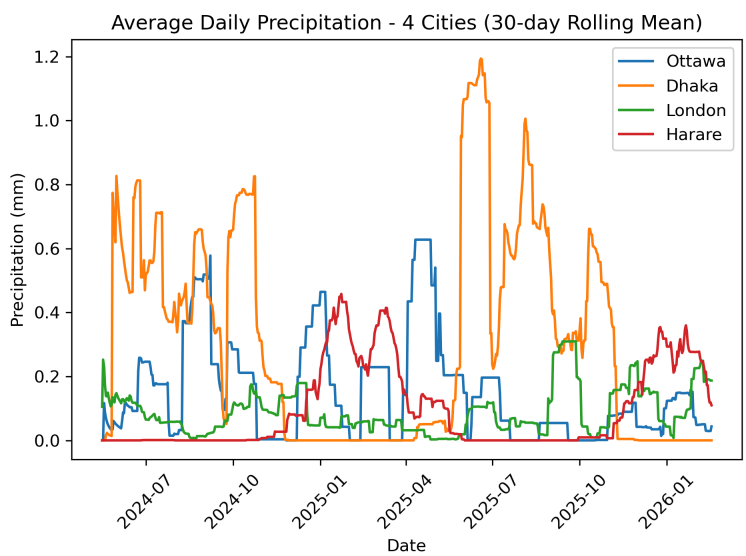
Precipitation behaves very differently from temperature. Unlike temperature, rainfall does not change smoothly over time. Instead, it shows irregular and sudden changes.

Comparison Across Four Cities

The 30-day rolling mean precipitation plot for Ottawa, Dhaka, London, and Harare shows clear differences:

- Dhaka has strong rainfall peaks, especially during monsoon periods.
- Ottawa shows moderate rainfall with some seasonal increase.
- London has relatively steady but low precipitation throughout the year.
- Harare shows rainfall concentrated in certain months, with long dry periods.

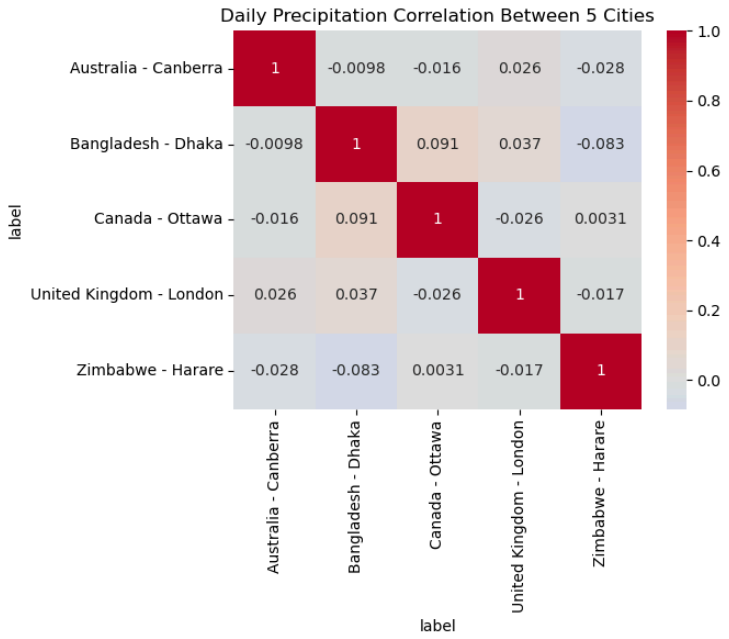
Unlike temperature, precipitation often stays close to zero and then suddenly increases. This explains the sharp spikes seen in the plot.



Correlation Between Cities

The precipitation correlation heatmap shows very weak correlations between cities. Most values are close to zero.

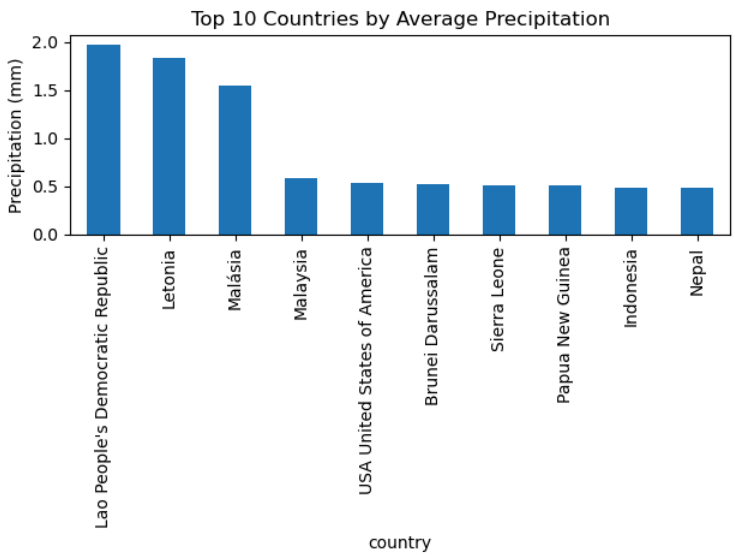
This means rainfall patterns are mostly independent across different regions. Unlike temperature, which follows global seasonal cycles, precipitation depends more on local climate systems, storms, and regional weather conditions.



Top Countries by Average Precipitation

The bar chart of the top 10 countries by average precipitation shows that some countries receive significantly higher rainfall than others. Countries in tropical or monsoon regions tend to have higher average precipitation compared to temperate regions.

This highlights how geographic location strongly affects rainfall patterns.



Overall, from the precipitation analysis, we observe:

- Rainfall is irregular and spiky
- Many days have little or no precipitation
- Strong regional differences exist
- Correlation between cities is generally weak

These results show that precipitation is more unpredictable than temperature. This suggests that forecasting precipitation may require different modeling approaches and careful handling of extreme values.

3.4 Correlation Analysis

A correlation heatmap was created to examine the relationships between the main numerical weather and air quality variables. Duplicate unit variables were removed (for example, only Celsius was kept instead of both Celsius and Fahrenheit) to avoid artificial correlations.

Key Observations

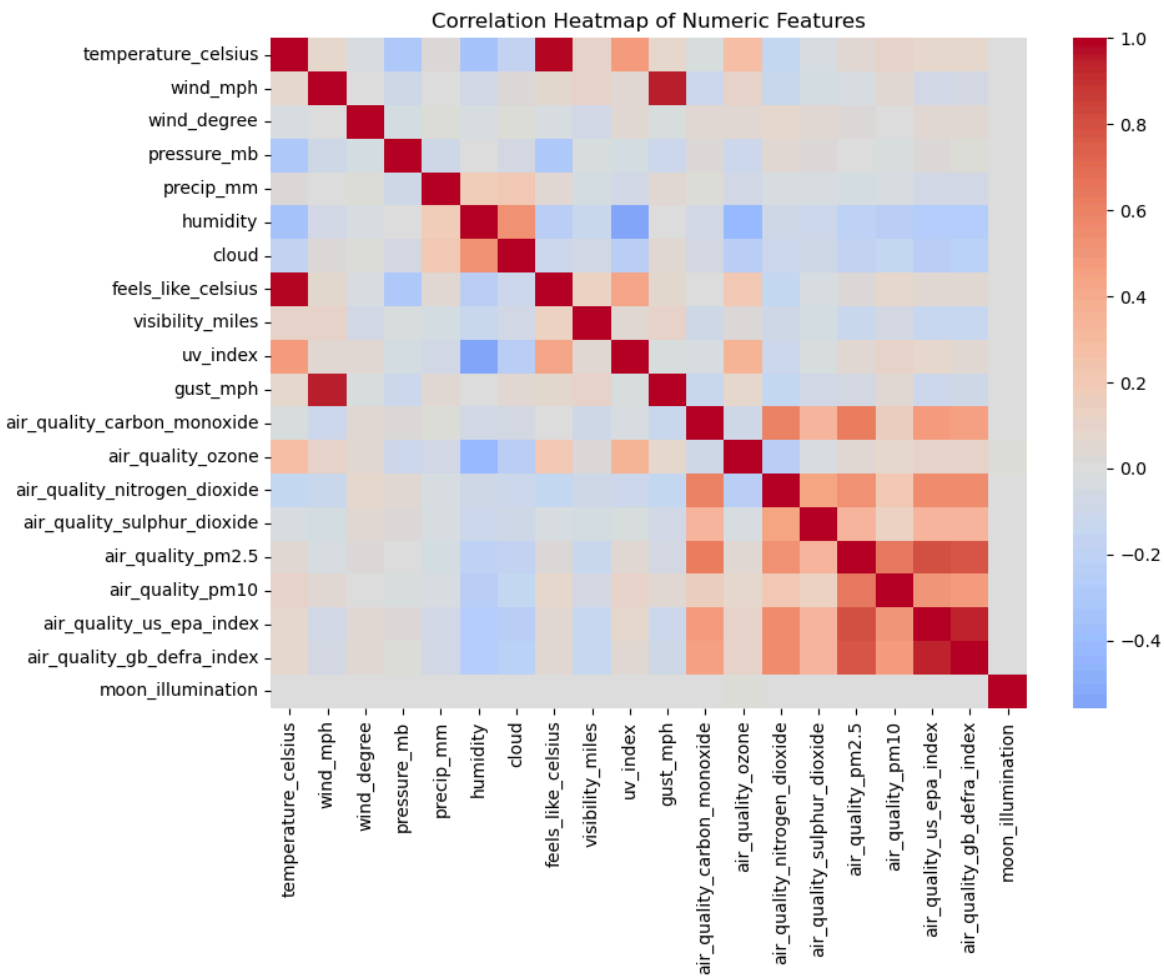
- There is a very strong positive correlation between **temperature** and **feels-like temperature**. This is expected because the feels-like value is calculated using temperature and other related factors.

- **Air quality variables** such as PM2.5, PM10, nitrogen dioxide, sulphur dioxide, and the EPA and DEFRA indices show strong positive correlations with each other. This indicates that these pollution measurements tend to increase and decrease together.

- There is a moderate positive relationship between **temperature and UV index**, which makes sense since sunnier days are often warmer.

- **Humidity and cloud cover** also show a positive relationship, which reflects typical weather patterns.

- Most other weather variables show weak to moderate correlations. This suggests that many environmental factors interact in complex ways rather than being strongly dependent on one another.



3.5 Summary of EDA Findings

The exploratory analysis shows several important patterns in the data:

- Clear seasonal temperature cycles, especially in cities with strong summer and winter differences
- Opposite seasonal trends between Northern and Southern Hemisphere cities
- Irregular and spiky precipitation patterns, with large differences across regions
- Strong relationships among air quality variables
- Mostly weak to moderate relationships among other weather features

Overall, the dataset shows realistic climate behavior and clear time-based patterns. These findings confirm that the data is suitable for time-series forecasting and further modeling.

4. Model building and evaluation

4.1 Forecasting setup

The goal was to forecast **daily average temperature (°C)** for a **single city (Ottawa, Canada)** using the *last_updated* timestamp as the time axis. The raw data contains multiple records per day, so the series was converted into a **daily temperature time series** by flooring *last_updated* to day-level and taking the **daily mean**. A complete daily date index was created, and missing days were filled using **interpolation** (reasonable for temperature, which changes smoothly day-to-day). The interpolation was performed only within the available historical data and did not use any information from the test period when constructing training features. This ensured that no data leakage occurred.

A **time-based split** was used to avoid leakage: the first **80%** of days were used for training and the last **20%** were used for testing. Performance was evaluated using **MAE, RMSE, and sMAPE**, and results were compared on the same test horizon.

4.2 The Chosen Models and Features

Univariate models (temperature-only)

These models use only the historical temperature series:

- **ETS (Exponential Smoothing)**: a strong baseline for smooth series with level/trend components. It is simple, fast, and interpretable.

- **SARIMA**: a classical time-series model that can capture autoregressive structure and seasonal effects.

These are included mainly as traditional baselines and to demonstrate classical forecasting.

Multivariate models (temperature + other weather signals)

Weather variables are physically related. Temperature is influenced by pressure systems, cloud cover, humidity, wind, precipitation, and UV. So multivariate modeling is expected to perform better than univariate-only models.

Two tree-based regression models were used:

- **XGBoost (multivariate)**: strong for nonlinear relationships and interactions, robust with limited data, and usually performs well with engineered lag features.

- **LightGBM (multivariate)**: similar strengths to XGBoost, often faster and sometimes more accurate depending on the dataset.

Feature construction (how multivariate forecasting was done)

Because XGBoost/LightGBM are not "time-series models" by default, the time-series problem was converted into supervised learning by creating **lagged features**:

- **Calendar features (*dayofweek, month, dayofyear*)** were included to capture recurring seasonal patterns. Temperature exhibits strong annual seasonality and weaker weekly structure. These variables help the model learn periodic behavior without explicitly modeling seasonality as in SARIMA.

- **Temperature lags (1, 2, 3, 7, 14 days)** were used to capture short-term autocorrelation and weekly memory. Temperature is highly dependent on recent values, and lag features allow the model to learn persistence and gradual transitions between weather regimes.

- **Exogenous variable lags (humidity, pressure, wind, cloud, precipitation, UV, visibility)** were included because these variables are physically related to temperature dynamics. For example, pressure systems influence air mass changes, cloud cover affects radiation balance, and wind patterns impact heat distribution. Using their lagged values allows the model to capture delayed effects while avoiding leakage.

- **Rolling statistics (7-day mean and standard deviation)** summarize recent trends and volatility. The rolling mean captures short-term warming or cooling trends, while the rolling standard deviation reflects stability versus variability in recent weather patterns.

Importantly, only **past observations** were used when constructing lag and rolling features. This ensured a strictly causal forecasting setup and prevented data leakage from future values.

4.3 Ensemble models

Two ensembles were constructed:

Univariate ensemble

Combined:

- ETS
- SARIMA
- XGBoost (univariate)

Multivariate ensemble

Combined:

- XGBoost (multivariate)
- LightGBM (multivariate)

To determine the contribution of each model, a validation subset was created from the last portion of the training data. Each model was first trained on the earlier part of the training set and evaluated on this validation slice. The validation errors (RMSE) were then used to compute inverse-error weights, meaning that models with lower validation error received higher weight in the ensemble.

After determining the weights, each model was retrained on the full training dataset, and final test forecasts were generated. The ensemble prediction was obtained as a **weighted average of the individual model forecasts**.

This approach improves robustness by leveraging the strengths of multiple models while reducing the impact of weaker individual forecasts.

4.4 Hyperparameter Tuning Strategy

To improve predictive performance while preserving the integrity of the forecasting setup, hyperparameter tuning was conducted using a time-aware validation approach. Instead of random cross-validation, the training data was split chronologically, where the final portion of the training period served as a validation subset. Each candidate model configuration was trained on the earlier training segment and evaluated on the validation subset using RMSE as the primary selection criterion. The best-performing configuration was then retrained on the full training dataset before final evaluation on the

test set. This approach ensured that no future information was used during model selection, thereby preventing data leakage.

ETS (Exponential Smoothing)

For ETS, tuning focused on structural components rather than numerical hyperparameters. Different combinations of trend and seasonality were evaluated, including additive trend and additive seasonal components. Weekly seasonality (*seasonal_periods* = 7) was tested given the daily resolution of the data. The configuration with the lowest validation RMSE was selected.

SARIMA

For SARIMA, model orders were optimized using an automated search procedure (*auto_arima*), constrained within reasonable ranges for autoregressive and moving average terms to prevent overfitting. Both seasonal and non-seasonal specifications were evaluated, with weekly seasonality (m = 7) considered. The model achieving the best validation performance was retained.

XGBoost (Univariate and Multivariate)

For XGBoost models, key hyperparameters were tuned, including:

- number of trees (*n_estimators*)
- learning rate
- maximum tree depth
- subsampling ratio
- column sampling ratio

Lower learning rates combined with a larger number of trees were tested to improve generalization. The optimal configuration was determined based on validation RMSE.

LightGBM (Multivariate)

For LightGBM, tuning focused on:

- number of leaves
- tree depth
- learning rate

- number of boosting iterations
- subsampling parameters

As with XGBoost, the configuration yielding the lowest validation error was selected.

Ensemble Models

After tuning individual models, ensemble weights were determined using validation errors. Inverse-RMSE weighting was applied so that models with lower validation error contributed more heavily to the final ensemble forecast. Final ensemble predictions were generated by applying these weights to the tuned model outputs on the test set.

This structured tuning process ensured fair comparison across models while maintaining chronological integrity and preventing information leakage.

4.4 Results and best model selection

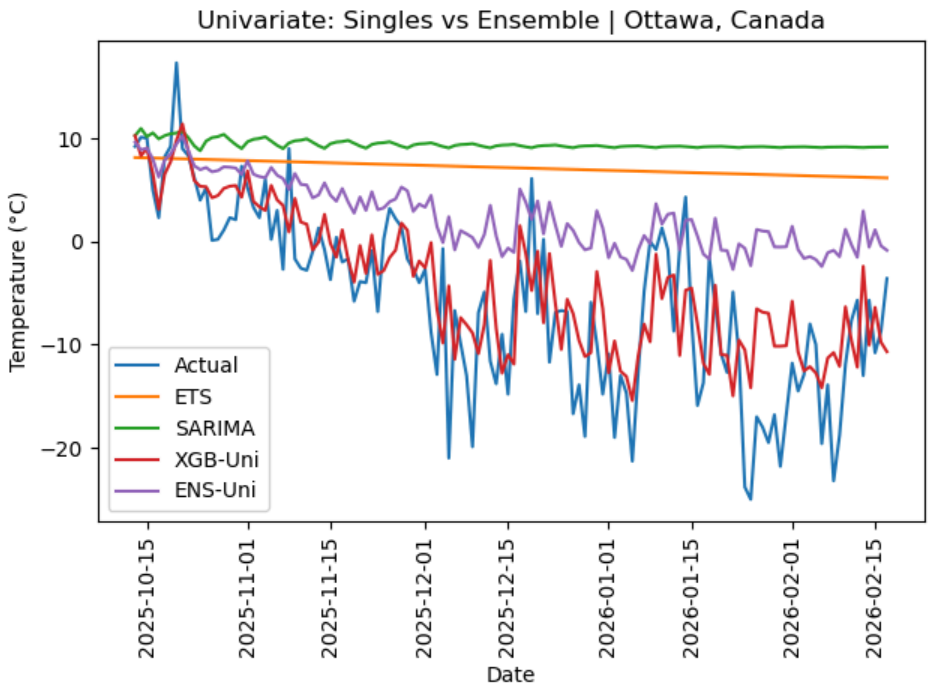
Univariate: single models vs ensemble

From the notebook outputs, univariate performance was:

- XGBoost (univariate) achieved the best univariate result (RMSE ≈ 5.86)
- ETS and SARIMA were substantially worse on this city/period
- Because ETS and SARIMA were weak, the univariate ensemble did not improve over XGBoost alone (ensemble RMSE was higher)

Interpretation: When some ensemble members are clearly poor, averaging can drag performance down.

Model	MAE	RMSE	sMAPE(%)
ETS	13.067339	15.232228	167.101618
SARIMA	15.146741	17.253714	169.701211
XGB-Uni	4.654062	5.861930	82.624953
ENS-Uni	8.551224	10.459465	150.047432



This figure presents the univariate forecasting results for daily average temperature in Ottawa, Canada, over the test period.

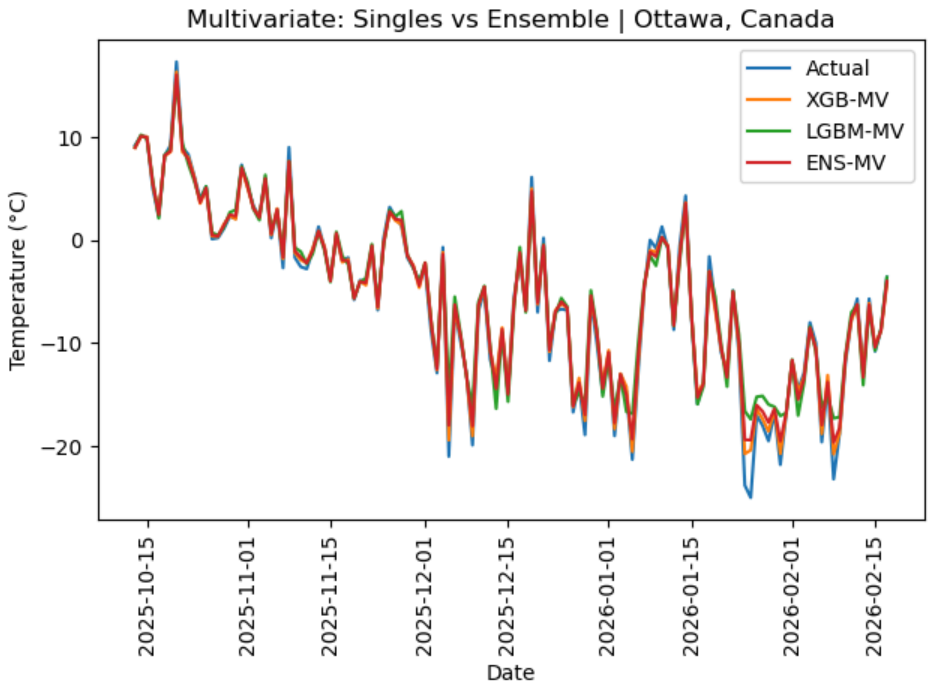
Multivariate: single models vs ensemble

Multivariate performance was much stronger overall:

- Ensemble (multivariate) was the best model overall (RMSE ≈ 0.43, lowest MAE and sMAPE as well)
- XGBoost multivariate was second-best
- The multivariate XGBoost was competitive but did not beat Ensemble multivariate in this run.

Model	MAE	RMSE	sMAPE(%)
XGB-MV	0.507448	0.761475	17.130575
LGBM-MV	1.075205	1.730395	25.500667

ENS-MV	0.436114	0.629672	16.072727



This figure presents the multivariate forecasting results for daily average temperature in Ottawa, Canada, over the test period.

Best model in this analysis (based on RMSE/MAE/sMAPE): Ensemble Multivariate

Univariate vs multivariate conclusion

Multivariate models were clearly better than univariate models in this analysis. The multivariate approach improved accuracy because it leveraged additional weather signals that contain predictive information about temperature dynamics (pressure, humidity, wind, cloud, precipitation, UV, etc.). Univariate ETS/SARIMA struggled because temperature does not always follow a stable seasonal pattern over a relatively short window (~540 days), and weather regimes can shift nonlinearly.

Final takeaway

- Best single model: Ensemble Multivariate
- Multivariate beats univariate in this experiment by a large margin.
- Ensembles are useful for robustness, but they only win when the member models are all reasonably strong. Here, the weak univariate models hurt the univariate ensemble.